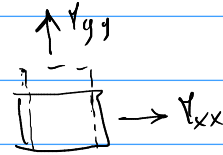


$$e_{xx} = \frac{1}{E} [\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})]$$



$$e_{xx} = \frac{1}{E} \epsilon_{xx} - \frac{\nu}{E} (\epsilon_{yy} + \epsilon_{zz})$$

$$\sigma_{ij} = \lambda e_{\alpha\alpha} \delta_{ij} + 2\mu e_{ij}$$

$$\left[\sigma_{ii} = \lambda e_{\alpha\alpha} \cdot \overset{3}{\delta_{ii}} + 2\mu \cdot e_{\alpha\alpha} = (3\lambda + 2\mu) \cdot e_{\alpha\alpha} \Rightarrow e_{\alpha\alpha} = \frac{\sigma_{\alpha\alpha}}{(3\lambda + 2\mu)} \right] \quad \text{deformação volumétrica}$$

$$\sigma_{ij} = \lambda \frac{\sigma_{\alpha\alpha}}{(3\lambda + 2\mu)} \cdot \delta_{ij} + 2\mu \cdot e_{ij} \Rightarrow$$

$$e_{ij} = \frac{1}{2\mu} \left(\sigma_{ij} - \lambda \frac{\sigma_{\alpha\alpha} \delta_{ij}}{3\lambda + 2\mu} \right) \quad (\lambda, \mu) \rightarrow (E, \nu)$$

$$\left[\lambda = \frac{2G\nu}{1-2\nu} = \frac{G(E-2G)}{3G-E} = K - \frac{2}{3}G = \frac{E\nu}{(1+\nu)(1-2\nu)} \right]$$

$$\left[G = \frac{\lambda(1-2\nu)}{2\nu} = \frac{3}{2}(K-\lambda) = \frac{E}{2(1+\nu)} \right]$$

$$e_{ij} = \frac{1}{E} \left[\sigma_{ij} - \nu (\sigma_{\alpha\alpha} \delta_{ij} - \sigma_{ij}) \right]$$

$$i \neq j \Rightarrow \delta_{ij} = 0 \Rightarrow e_{ij} = \frac{1+\nu}{E} \sigma_{ij} \quad \left\{ \begin{array}{l} \vdots \\ \vdots \\ \vdots \end{array} \right.$$

$$i = j \Rightarrow \delta_{ij} = 1 \Rightarrow e_{ij} = \frac{1}{E} \left[\sigma_{ij} - \nu (\sigma_{\alpha\alpha} - \sigma_{ij}) \right] \quad \left\{ \begin{array}{l} \vdots \\ \vdots \\ \vdots \end{array} \right.$$